

R Programs Analysis and Algorithms

1. Armstrong Number Checker (armstrong.R)

Algorithm:

1. Define a function `armstrong()` that takes a number as input
2. Split the number into individual digits
3. Calculate the sum of each digit raised to the power of the total number of digits
4. Return TRUE if the sum equals the original number, FALSE otherwise
5. Get a number from user input
6. Check if it's an Armstrong number and display the result

Purpose: Checks if a number is an Armstrong number (a number that equals the sum of its digits each raised to the power of the number of digits).

2. Covariance Calculator (Covariance.R)

Algorithm:

1. Load a dataset from a CSV file
2. Display the first few rows to understand the data structure
3. Select two columns from the dataset
4. Calculate the covariance between the two columns
5. Calculate the correlation between the two columns
6. Display the results

Purpose: Calculates and displays the covariance and correlation between two columns in a dataset.

3. Decision Tree Classifier (Decison Tree.R)

Algorithm:

1. Load required libraries (rpart, rpart.plot)
2. Load the iris dataset
3. Build a decision tree model to classify iris species based on sepal and petal measurements
4. Display a summary of the model
5. Plot the decision tree visually
6. Make predictions using the model
7. Calculate and display the model's accuracy

Purpose: Creates a decision tree classification model for the iris dataset and evaluates its performance.

4. Factorial Calculator (Recursive) (factorial.R)

Algorithm:

1. Get a number from user input
2. Define a recursive function `fact1()` to calculate factorial:
 - If the number is 0, return 1
 - Otherwise, return the number multiplied by the factorial of (number-1)
3. Call the function with the user's input
4. Display the result

Purpose: Calculates the factorial of a number using recursion.

5. Factorial Calculator (Iterative) (factorial.R)

Algorithm:

1. Get a number from user input
2. Define a function `factorial()` that:
 - Initializes the result to 1
 - Iterates from the input number down to 1, multiplying the result by each number
 - Returns the final product
3. Call the function with the user's input
4. Display the result

Purpose: Calculates the factorial of a number using iteration.

6. Fibonacci Series Generator (fibnocci.R)

Algorithm:

1. Define a function `fib()` that generates a Fibonacci series of n terms:
 - Initialize the series with [0, 1]
 - For each subsequent position, add the previous two numbers
 - Return the complete series
2. Get the desired number of terms from user input
3. Call the function to generate the series
4. Display the Fibonacci series

Purpose: Generates and displays a Fibonacci series up to n terms.

7. K-Means Clustering (KMEAN.R)

Algorithm:

1. Load required libraries (ggplot2)
2. Load the iris dataset
3. Remove any rows with missing values and select only numeric columns
4. Set a random seed for reproducibility
5. Apply K-means clustering with 3 centers
6. Add cluster assignments to the dataset
7. Extract the centroids of each cluster
8. Create a scatter plot showing:
 - Data points colored by cluster assignment
 - Cluster centroids as larger blue points
9. Display the plot

Purpose: Performs K-means clustering on the iris dataset and visualizes the results.

8. Linear Regression (linear.R)

Algorithm:

1. Load required libraries (caTools)
2. Create a sample dataset with length and weight variables
3. Split the data into training (70%) and test (30%) sets
4. Build a linear regression model predicting weight based on length
5. Display a summary of the model
6. Make predictions on the test set
7. Calculate the mean squared error
8. Create a scatter plot of the data with the regression line
9. Display the plot

Purpose: Builds a linear regression model and visualizes the relationship between two variables.

9. Logistic Regression (Logistic Regression.R)

Algorithm:

1. Load required libraries (caTools)
2. Load and preprocess the Titanic dataset

3. Convert categorical variables to factors
4. Split the data into training (80%) and test (20%) sets
5. Build a logistic regression model predicting survival based on various features
6. Make predictions on the test set
7. Calculate the model's accuracy
8. Create a plot showing:
 - The sigmoid curve
 - Predicted data points colored by class (survived/not survived)
9. Display the plot

Purpose: Builds a logistic regression model to predict survival on the Titanic and visualizes the results.

10. Statistical Calculations (NCR.R)

Algorithm:

1. Define custom functions for statistical calculations (min, max, mean, median, variance, standard deviation, mode)
2. Load the iris dataset
3. Select only numeric columns
4. For each numeric column:
 - Calculate all statistical measures
 - Display the results

Purpose: Calculates and displays various statistical measures for numeric columns in a dataset.

11. Palindrome Checker (Number) (paliandrome.R)

Algorithm:

1. Get a number from user input
2. Initialize a variable to store the reversed number
3. While the number is greater than 0:
 - Extract the last digit
 - Add it to the reversed number (multiplied by 10)
 - Remove the last digit from the original number
4. Check if the reversed number equals the original number
5. Display whether the number is a palindrome or not

Purpose: Checks if a number reads the same backward as forward.

12. Prime Number Checker (prime or not.R)

Algorithm:

1. Get a number from user input
2. Define a function `prime()` that:
 - Returns FALSE if the number is less than or equal to 1
 - Checks if the number is divisible by any integer from 2 to (number-1)
 - Returns TRUE if no divisors are found
3. Call the function with the user's input
4. Display whether the number is prime or not

Purpose: Determines if a given number is prime.

13. Statistical Calculations (statistics.R)

Algorithm:

1. Define custom functions for statistical calculations (min, max, mean, median, variance, standard deviation, mode)
2. Load the iris dataset
3. Select only numeric columns
4. For each numeric column:
 - Calculate all statistical measures
 - Display the results

Purpose: Calculates and displays various statistical measures for numeric columns in a dataset (duplicate of NCR.R).

14. Palindrome Checker (String) (string paliandrome.R)

Algorithm:

1. Define a function that:
 - Gets a string from user input
 - Splits the string into individual characters
 - Reverses the character array
 - Joins the reversed array back into a string
 - Compares the reversed string with the original
 - Displays whether the string is a palindrome or not

2. Call the function

Purpose: Checks if a string reads the same backward as forward.

15. Strong Number Checker (strong or not.R)

Algorithm:

1. Get a number from user input
2. Define a function `fact()` to calculate factorial iteratively
3. Initialize a variable to store the sum of factorial of digits
4. For each digit in the number:
 - Calculate its factorial
 - Add it to the sum
5. Check if the sum equals the original number
6. Display whether the number is a strong number or not

Purpose: Checks if a number is a strong number (the sum of the factorials of its digits equals the number itself).

16. Support Vector Machine Classifier (SVM.R)

Algorithm:

1. Load required libraries (e1071, ggplot2)
2. Load the iris dataset
3. Build an SVM model to classify iris species based on sepal measurements
4. Create a grid of points covering the feature space
5. Use the model to predict the class for each point in the grid
6. Create a plot showing:
 - The decision boundaries as colored regions
 - The original data points
 - Support vectors highlighted with special markers
7. Display the plot

Purpose: Creates a Support Vector Machine classification model for the iris dataset and visualizes the decision boundaries.